

# AgrilIntelligence: An Explainable AI-Based Agricultural Decision Support System

## Project Review-1



### Under the Guidance of

**Dr N Sunanda**

Assistant Professor,

**Department of Computer Science and Engineering,**

KLH CSE Bowrampet Campus



### Team Members

**Meghna- 2420030401**

**Thanusree – 2420030442**

**Rasagna - 2420030117**

# Outline of Research Work



# Problem Statement

Farmers must make high-stakes crop decisions with fragmented, incomplete information — navigating unpredictable weather, shifting soil conditions, and volatile market prices, often without integrated guidance.

- **Current Situation :**  
Crop selection relies on traditional knowledge. Weather is unpredictable, and market prices fluctuate often causing financial losses.
- **Information Gap :**  
Existing systems address crop, weather, or market data in isolation. Datasets are collected at different intervals, making integration difficult.
- **Farmer Risk :**  
Without unified intelligence, farmers face compounded uncertainty — choosing crops without full visibility into environmental suitability *and* economic viability..

**"Farmers need an adaptive decision-support system that integrates crop/soil, weather, and market information to recommend suitable crops and support better yield and profitability decisions."**

# Objectives

➤ **Unified Data Integration**

Integrate crop/soil, weather, and market-price datasets into one cohesive system.

➤ **Crop Suitability Prediction**

Predict suitable crops based on soil properties and environmental conditions.

➤ **Weather Impact Analysis**

Analyze how weather patterns influence crop growth and expected yield.

➤ **Market Price Forecasting**

Forecast market-price trends to support economically profitable crop selection.

➤ **Multi-Rate Data Handling**

Align datasets collected at different intervals through preprocessing and synchronization.

➤ **Farmer-Friendly Interface**

Deliver clear, actionable recommendations through an intuitive decision-support dashboard.

# Literature Survey

| Study / Research Area               | Data Used                                      | Method / Model                                  | Major Finding                                   | Limitation                                                          |
|-------------------------------------|------------------------------------------------|-------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------|
| Crop Yield Prediction (ML)          | Crop/soil + weather data                       | Random Forest, Gradient Boosting, Deep Learning | Strong yield prediction performance             | Focuses mainly on yield, not economics                              |
| Agricultural Price Prediction       | Historical prices + weather/market variables   | Regression, XGBoost, Time-Series models         | Weather and market data improve price forecasts | Lacks detailed crop suitability integration                         |
| Multi-Modal Agricultural Prediction | Crop/yield + remote sensing + weather + market | Ensemble / Deep Learning                        | Connects productivity and market information    | Data synchronization and regional generalization remain challenging |

# Research Gap Identification

## ☐ **Data Synchronization**

Weather, crop, and market data follow different reporting frequencies, making unified analysis difficult.

## ☐ **Regional Adaptability**

Models trained for one region often fail elsewhere due to differing soil, climate, and market conditions.

## ☐ **Unexpected Market Events**

Policy shifts, export restrictions, and fuel-price changes create sudden disruptions traditional models miss.

## ☐ **Post-Harvest Factors**

Most systems focus on production but overlook storage, transportation, and crop perishability.

# Innovation, Creativity, and Novelty

## ✓ Innovation

Integration of three critical agricultural data sources — **crop/soil, weather, and market price** — into a single adaptive system.

## ✓ Creativity

Complex multi-source agricultural data is distilled into **simple, actionable recommendations** any farmer can understand and act on.

## ✓ Novelty

Instead of asking "*Which crop can grow?*", the system answers: "**Which crop is suitable *and* economically promising under current conditions?**"

❖ *Future directions may include: news/policy event analysis, regional model adaptation, and post-harvest logistics integration.*

# Feasibility

- **Technical Stack**

Python, Pandas, Scikit-learn, ML models, PostgreSQL/Firebase, REST APIs, and a web dashboard for visualization.

- **Data Sources**

Public agricultural datasets, historical crop/yield records, weather APIs, and agricultural market-price databases.

- **Operational Design**

Dashboard-based recommendations designed for easy interpretation. Adaptable to specific regional contexts.

- **Economic Viability**

Built on open-source software and publicly available datasets. No expensive hardware required for the initial prototype

# Data Science Flow Diagram – Adaptive Smart Farming Decision Support System

